

Zadania

0.1 Wyświetl wszystkie informacje użytkownika Alice.

```
curl -X 'GET' 'http://127.0.0.1:5000/user/alice' -H accept: '/' | jq
```

```
[user@parrot]~$ curl -X 'GET' 'http://127.0.0.1:5000/user/alice' -H accept: '*'/*' | jq
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total     Spent    Left     Speed
100    116  100    116    0     0  21678      0 --:--:-- --:--:-- --:--:-- 23200
{
  "age": 25,
  "city": "Warsaw",
  "email": "alice@example.com",
  "hobbies": [
    "reading",
    "chess",
    "cycling"
  ],
  "name": "Alice Smith"
}
```

0.2 Wyświetl imię i nazwisko użytkownika Alice.

```
curl -X 'GET' 'http://127.0.0.1:5000/user/alice' -H accept: '/' | jq -r ".name"
```

```
[user@parrot]~$ curl -X 'GET' 'http://127.0.0.1:5000/user/alice' -H accept: '*'/*' | jq -r ".name"
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total     Spent    Left     Speed
100    116  100    116    0     0  11610      0 --:--:-- --:--:-- --:--:-- 12888
Alice Smith
```

0.3 Wyświetl adres e-mail użytkownika Bob.

```
curl -X 'GET' 'http://127.0.0.1:5000/user/bob' -H accept: '/' | jq -r ".email"
```

```
[user@parrot]~$ curl -X 'GET' 'http://127.0.0.1:5000/user/bob' -H accept: '*'/*' | jq -r ".email"
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total     Spent    Left     Speed
100    105  100    105    0     0  53272      0 --:--:-- --:--:-- --:--:-- 102k
bob@example.com
```

0.4 Wyświetl wiek użytkownika Alice.

```
curl -X 'GET' 'http://127.0.0.1:5000/user/alice' -H accept: '/' | jq -r ".age"
```

```
[user@parrot]~$ curl -X 'GET' 'http://127.0.0.1:5000/user/alice' -H accept: '*/*' | jq -r ".age"
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total   Spent    Left  Speed
100    116    100    116     0     0  29997       0 --:--:-- --:--:-- --:--:-- 38666
25
```

0.5 Wyświetl nazwę miasta, w którym mieszka użytkownik Bob.

```
curl -X 'GET' 'http://127.0.0.1:5000/user/bob' -H accept: '/' | jq -r ".city"
```

```
[user@parrot]~$ curl -X 'GET' 'http://127.0.0.1:5000/user/bob' -H accept: '*/*' | jq -r ".city"
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total   Spent    Left  Speed
100    105    100    105     0     0  32387       0 --:--:-- --:--:-- --:--:-- 35000
Krakow
```

0.6 Wyświetl hobby użytkownika Alice.

```
[user@parrot]~$ curl -X 'GET' 'http://localhost:5000/user/alice' -H 'accept: */*' | jq ".hobbies"
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total   Spent    Left  Speed
100    116    100    116     0     0  49871       0 --:--:-- --:--:-- --:--:-- 58000
[
  "reading",
  "chess",
  "cycling"
]
```

0.7 Sprawdź, czy jednym z hobby użytkownika Bob są gry.

```
[user@parrot]~$ curl -X 'GET' 'http://localhost:5000/user/bob' -H 'accept: */*' | jq -r 'any(.hobbies[]; . == "games")'
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total   Spent    Left  Speed
100    105    100    105     0     0  56695       0 --:--:-- --:--:-- --:--:-- 102k
true
```

0.8 Wyświetl pierwsze hobby użytkownika Alice.

```
[user@parrot]~$ curl -X 'GET' 'http://localhost:5000/user/alice' -H 'accept: */*' | jq -r '.hobbies[0]'
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total   Spent    Left  Speed
100    116    100    116     0     0  39876       0 --:--:-- --:--:-- --:--:-- 58000
reading
```

0.9 Sprawdź, ile hobby ma użytkownik Bob, a ile użytkownik Alice.

```
[user@parrot]~$ curl -X 'GET' 'http://localhost:5000/user/bob' -H 'accept: */*' | jq -r '.hobbies | length'
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total   Spent    Left  Speed
100    105    100    105     0     0   9110       0 --:--:-- --:--:-- --:--:-- 9545
2
```

```
[user@parrot]-[~]
$ curl -X 'GET' 'http://localhost:5000/user/alice' -H 'accept: */*' | jq -r '.hobbies | length'
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload    Total      Spent    Left  Speed
100    116    100    116      0      0   61052      0 --:--:-- --:--:-- --:--:--  113k
3
```

0.10 Wyświetl nazwę użytkownika i miasto jako jeden ciąg znaków, np. "Alice Smith (Warsaw)".

Zmodyfikuj polecenie, aby tekst był wyświetlany jako "Alice Smith from Warsaw".

```
[user@parrot]-[~]
$ curl -X 'GET' 'http://localhost:5000/user/alice' -H 'accept: */*' | jq -r '"\(.name) (\(.city))"'
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload    Total      Spent    Left  Speed
100    116    100    116      0      0   46104      0 --:--:-- --:--:-- --:--:--  58000
Alice Smith (Warsaw)
```

```
[user@parrot]-[~]
$ curl -X 'GET' 'http://localhost:5000/user/alice' -H 'accept: */*' | jq -r '"\(.name) from \(.city)"'
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload    Total      Spent    Left  Speed
100    116    100    116      0      0   14640      0 --:--:-- --:--:-- --:--:--  16571
Alice Smith from Warsaw
```

0.11 Wyświetl wszystkie przedmioty. Wyświetl nazwy wszystkich przedmiotów.

```
[user@parrot]-[~]
$ curl -X 'GET' 'http://localhost:5000/items' -H 'accept: */*'
[{"id":1,"name":"itemX","price":15.99},{ "id":2,"name":"superItem","price":99.9},{ "id":3,"name":"item123","price":5.49},{ "id":4,"name":"itemY","price":29.0}]
```

```
[x]-[user@parrot]-[~]
$curl -X 'GET' 'http://localhost:5000/items' -H 'accept: */*' | jq -r
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total     Spent    Left     Speed
100    157    100    157     0     0  15461      0 --:--:-- --:--:-- --:--:-- 15700
[
  {
    "id": 1,
    "name": "itemX",
    "price": 15.99
  },
  {
    "id": 2,
    "name": "superItem",
    "price": 99.9
  },
  {
    "id": 3,
    "name": "item123",
    "price": 5.49
  },
  {
    "id": 4,
    "name": "itemY",
    "price": 29
  }
]
```

```
[user@parrot]-[~]
$curl -X 'GET' 'http://localhost:5000/items' -H 'accept: */*' | jq -r .[].name
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total     Spent    Left     Speed
100    157    100    157     0     0  79133      0 --:--:-- --:--:-- --:--:-- 153k
itemX
superItem
item123
itemY
```

0.12 Wyświetl przedmioty droższe niż 20 pln.

```
[x]-[user@parrot]-[~]
$curl -X 'GET' 'http://localhost:5000/items' -H 'accept: */*' | jq -r '[] | select(.price > 20.00)'
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total     Spent    Left     Speed
100    157    100    157     0      0  79252      0 --:--:-- --:--:-- --:--:--  153k
{
  "id": 2,
  "name": "superItem",
  "price": 99.9
}
{
  "id": 4,
  "name": "itemY",
  "price": 29
}
```

0.13 Wyświetl przedmioty tańsze niż 30 pln.

```
[user@parrot]-[~]
$curl -X 'GET' 'http://localhost:5000/items' -H 'accept: */*' | jq -r '[] | select(.price < 30.00)'
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total     Spent    Left     Speed
100    157    100    157     0      0  42697      0 --:--:-- --:--:-- --:--:--  52333
{
  "id": 1,
  "name": "itemX",
  "price": 15.99
}
{
  "id": 3,
  "name": "item123",
  "price": 5.49
}
{
  "id": 4,
  "name": "itemY",
  "price": 29
}
```

0.14 Posortuj przedmioty według ceny rosnąco, a następnie malejąco.

```
[x]-[user@parrot]-[~]
$curl -X 'GET' 'http://localhost:5000/items' -H 'accept: */*' | jq -r 'sort_by(.price)'
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total   Spent    Left  Speed
100    157  100    157    0     0  72150      0 --:--:-- --:--:-- --:--:--  78500
[
  {
    "id": 3,
    "name": "item123",
    "price": 5.49
  },
  {
    "id": 1,
    "name": "itemX",
    "price": 15.99
  },
  {
    "id": 4,
    "name": "itemY",
    "price": 29
  },
  {
    "id": 2,
    "name": "superItem",
    "price": 99.9
  }
]
```

```
[x]-[user@parrot]-[~]
$curl -X 'GET' 'http://localhost:5000/items' -H 'accept: */*' | jq -r 'sort_by(.price) | reverse'
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total   Spent    Left  Speed
100    157  100    157    0     0  86263      0 --:--:-- --:--:-- --:--:--  153k
[
  {
    "id": 2,
    "name": "superItem",
    "price": 99.9
  },
  {
    "id": 4,
    "name": "itemY",
    "price": 29
  },
  {
    "id": 1,
    "name": "itemX",
    "price": 15.99
  },
  {
    "id": 3,
    "name": "item123",
    "price": 5.49
  }
]
```

0.15 Pobierz sumę cen wszystkich przedmiotów.

```
[x]-[user@parrot]-[~]
$curl -X 'GET' 'http://localhost:5000/items' -H 'accept: */*' | jq -r 'map(.price) | add'
```

% Total	% Received	% Xferd	Average Speed		Time	Time	Time	Current
			Dload	Upload	Total	Spent	Left	Speed
100	157	100	157	0	0	48727	0	--:--:-- --:--:-- --:--:-- 52333
150.38								

0.16 Wyświetl przedmioty, których nazwa zawiera "item".

```
[user@parrot]-[~]
$curl -X 'GET' 'http://localhost:5000/items' -H 'accept: */*' | jq -r '.[ ] | select(.name | test("item"))'
```

% Total	% Received	% Xferd	Average Speed		Time	Time	Time	Current
			Dload	Upload	Total	Spent	Left	Speed
100	157	100	157	0	0	60547	0	--:--:-- --:--:-- --:--:-- 78500

```
{
  "id": 1,
  "name": "itemX",
  "price": 15.99
}
{
  "id": 3,
  "name": "item123",
  "price": 5.49
}
{
  "id": 4,
  "name": "itemY",
  "price": 29
}
```

0.17 Wyświetl przedmioty w przedziale cenowym 10-30 pln.

```
[user@parrot]-[~]
$curl -X 'GET' 'http://localhost:5000/items' -H 'accept: */*' | jq -r 'map(select(.price >= 10.00 and .price <= 30.00))'
```

% Total	% Received	% Xferd	Average Speed		Time	Time	Time	Current
			Dload	Upload	Total	Spent	Left	Speed
100	157	100	157	0	0	67410	0	--:--:-- --:--:-- --:--:-- 78500

```
[
  {
    "id": 1,
    "name": "itemX",
    "price": 15.99
  },
  {
    "id": 4,
    "name": "itemY",
    "price": 29
  }
]
```

0.18 Wyświetl użytkowników, którzy mają więcej niż 2 hobby.

0.19 Wyślij do serwera request typu POST (endpoint /echo) i wyświetl odpowiedź.

```
[x]-[user@parrot]-[~]
$curl -X 'POST' 'http://localhost:5000/echo' -H 'accept: /' -H 'Content-Type: application/json' -d '{}'
```

```
{"received":{}}
```

0.20 Wyślij do serwera request typu POST, który doda nowy przedmiot (użyj endpointa /items). Wyświetl wszystkie przedmioty po dodaniu.

```
[x]-[user@parrot]-[~]
→ $curl -X 'POST' 'http://localhost:5000/echo' -H 'accept:/' -H 'Content-Type: application/json' -d '{}'
{"received":{}}
[user@parrot]-[~]
→ $curl -X 'POST' \
'http://localhost:5000/items' \
-H 'accept: */*' \
-H 'Content-Type: application/json' \
-d '{
  "name": "myItem",
  "price": 15
}'
{"id":5,"name":"myItem","price":15}
```

```
[user@parrot]-[~]
→ $curl -X 'GET' 'http://localhost:5000/items' -H 'accept: */*' | jq -r
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           %    0     0     0         0      0      0      0
100    193  100    193    0     0  86392      0 --:--:-- --:--:-- --:--:-- 96500
[
  {
    "id": 1,
    "name": "itemX",
    "price": 15.99
  },
  {
    "id": 2,
    "name": "superItem",
    "price": 99.9
  },
  {
    "id": 3,
    "name": "item123",
    "price": 5.49
  },
  {
    "id": 4,
    "name": "itemY",
    "price": 29
  },
  {
    "id": 5,
    "name": "myItem",
    "price": 15
  }
]
```

0.21 Za pomocą requestu typu PUT, aktualizuj cenę przedmiotu o numerze 1. Wyświetl wszystkie przedmioty.

```
[user@parrot]~  
$ curl -X 'PUT' 'http://localhost:5000/items/1' -H 'Content-Type: application/json' -d '{"price":90}'  
{"id":1,"name":"itemX","price":90}
```

```
[user@parrot]~  
$ curl -X 'GET' 'http://localhost:5000/items' -H 'accept: */*' | jq -r  
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current  
           Dload  Upload   Total      Spent    Left     Speed  
100    190    100    190     0     0    113k      0 --:--:-- --:--:-- --:--:-- 185k  
[  
  {  
    "id": 1,  
    "name": "itemX",  
    "price": 90  
  },  
  {  
    "id": 2,  
    "name": "superItem",  
    "price": 99.9  
  },  
  {  
    "id": 3,  
    "name": "item123",  
    "price": 5.49  
  },  
  {  
    "id": 4,  
    "name": "itemY",  
    "price": 29  
  },  
  {  
    "id": 5,  
    "name": "myItem",  
    "price": 15  
  }  
]
```

0.22 Za pomocą requestu typu DELETE, usuń przedmiot o numerze 3. Wyświetl wszystkie przedmioty.

```
[user@parrot]~$ curl -X 'DELETE' 'http://localhost:5000/items/3' -H 'accept: */*'
{"result": "Item deleted if existed"}
[user@parrot]~$ curl -X 'GET' 'http://localhost:5000/items' -H 'accept: */*' | jq -r
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total     Spent    Left     Speed
100    151    100    151     0     0  73229      0 --:--:-- --:--:-- --:--:-- 75500
[
  {
    "id": 1,
    "name": "itemX",
    "price": 90
  },
  {
    "id": 2,
    "name": "superItem",
    "price": 99.9
  },
  {
    "id": 4,
    "name": "itemY",
    "price": 29
  },
  {
    "id": 5,
    "name": "myItem",
    "price": 15
  }
]
```

0.23 Wyświetl przedmiot o największej cenie.

```
[user@parrot]~$ curl -X 'GET' 'http://localhost:5000/items' -H 'accept: */*' | jq -r 'sort_by(.price) | reverse
| .[0]'
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total     Spent    Left     Speed
100    151    100    151     0     0   97k      0 --:--:-- --:--:-- --:--:-- 147k
{
  "id": 2,
  "name": "superItem",
  "price": 99.9
}
```

0.24 Wyświetl wszystkie przedmioty z ceną podwyższoną o 10%.

```
[x]-[user@parrot]-[~]
$curl -X 'GET' 'http://localhost:5000/items' -H 'accept: */*' | jq -r '.[0] | .price=(.price*1.1)'
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total     Spent    Left  Speed
100    151    100    151     0     0  67470      0 --:--:-- --:--:-- --:--:--  75500
{
  "id": 1,
  "name": "itemX",
  "price": 99.000000000000001
}
{
  "id": 2,
  "name": "superItem",
  "price": 109.890000000000001
}
{
  "id": 4,
  "name": "itemY",
  "price": 31.900000000000002
}
{
  "id": 5,
  "name": "myItem",
  "price": 16.5
}
```

0.25 Pobierz nazwy wszystkich użytkowników i ich wiek.

0.26 Wyświetl hobby Alice jako string połączony przecinkami.

```
[user@parrot]-[~]
$curl -X 'GET' 'http://localhost:5000/user/alice' -H 'accept: */*' | jq '.hobbies | join(", ")'
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total     Spent    Left  Speed
100    116    100    116     0     0  82327      0 --:--:-- --:--:-- --:--:--  113k
"reading, chess, cycling"
```

0.27 Pobierz pierwszy i ostatni przedmiot z listy.

```
[user@parrot]-[~]
$curl -X 'GET' 'http://localhost:5000/items' -H 'accept: */*' | jq '.[0], .[-1]'
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total     Spent    Left  Speed
100    151    100    151     0     0  87232      0 --:--:-- --:--:-- --:--:--  147k
{
  "id": 1,
  "name": "itemX",
  "price": 90
}
{
  "id": 5,
  "name": "myItem",
  "price": 15
}
```

0.28 Sprawdź, czy są przedmioty droższe niż 100 pln.

```
[user@parrot]~]
$ curl -X 'GET' 'http://localhost:5000/items' -H 'accept: */*' | jq '[] | (.price > 100)'
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total     Spent    Left  Speed
100    151    100    151     0     0  70004      0 --:--:-- --:--:-- --:--:--  75500
false
false
false
false
```

0.30 Zaktualizuj przedmiot id=2. Wyświetl odpowiedź serwera i następnie listę wszystkich przedmiotów.

```
[user@parrot]~]
$ curl -X 'PUT' 'http://localhost:5000/items/2' -H 'Content-Type: application/json' -d '{"name": "dsadasda"}'
{"id":2,"name":"dsadasda","price":99.9}
[user@parrot]~]
$ curl -X 'GET' 'http://localhost:5000/items' -H 'accept:/' | jq -r
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total     Spent    Left  Speed
100    150    100    150     0     0  11451      0 --:--:-- --:--:-- --:--:--  11538
[
  {
    "id": 1,
    "name": "itemX",
    "price": 90
  },
  {
    "id": 2,
    "name": "dsadasda",
    "price": 99.9
  },
  {
    "id": 4,
    "name": "itemY",
    "price": 29
  },
  {
    "id": 5,
    "name": "myItem",
    "price": 15
  }
]
```